

Java OOP — Access Modifiers

Notes on private, default, protected, and public access.

1. What is an Access Modifier?

Access modifiers control where a class, variable, method, or constructor can be accessed from.

2. The Four Access Modifiers

Java commonly uses four access levels: **private**, **default/package-private**, **protected**, and **public**.

3. private

private is the most restrictive modifier. A private member can be accessed only inside its declaring class.

```
class Student {  
    private int marks;  
  
    void display() {  
        System.out.println(marks); // allowed  
    }  
}  
  
// s.marks = 80; // not allowed
```

4. public

public provides the widest access. A public member can be accessed from anywhere the class is accessible.

```
class Student {  
    public String name;  
}  
  
Student s = new Student();  
s.name = "Sanjay"; // allowed
```

5. default (Package-Private)

If no access modifier is written, the member has default/package-private access. It can be accessed by classes in the same package.

```
class Student {  
    String name; // default  
    int marks;   // default  
}
```

6. protected

protected allows access inside the same class and same package, and through inheritance from subclasses. Its main use becomes clearer with Inheritance.

```
class Student {  
    protected int marks;  
}
```

7. Easy Way to Remember

private → most restricted
default → same package
protected → same package + inheritance
public → most accessible

8. Why Encapsulation Uses private

Private fields prevent direct outside modification. Public methods can provide controlled access and validation.

```
class Employee {
    private double salary;

    public void setSalary(double salary) {
        if (salary >= 0) {
            this.salary = salary;
        }
    }

    public double getSalary() {
        return salary;
    }
}
```

9. Complete Example

```
class Student {
    private int a = 10;
    int b = 20;
    protected int c = 30;
    public int d = 40;

    void display() {
        System.out.println(a);
        System.out.println(b);
        System.out.println(c);
        System.out.println(d);
    }
}

public class AccessExample {
    public static void main(String[] args) {
        Student s = new Student();

        // System.out.println(s.a); // private - not allowed
        System.out.println(s.b);    // default
        System.out.println(s.c);    // protected
        System.out.println(s.d);    // public
    }
}
```

10. Important Interview Questions

Most restrictive? private.

Least restrictive? public.

No modifier? Default/package-private access.

Why protected? It is closely related to inheritance.

11. Top-Level Classes

A top-level class can generally be public or default/package-private. It cannot be private or protected.

```
public class Student {  
}  
  
class Employee {  
}  
  
// private class Test { } // not allowed  
// protected class Test { } // not allowed
```

12. Logic Building

If outside classes must not directly modify a variable → **private**. If a method should be widely accessible → **public**. Same package only → **default**. Access through inheritance → **protected**.

13. Practice Problem

Create a BankAccount class with private double balance, public deposit(), and public getBalance(). deposit() should add the amount only when amount > 0; otherwise print Invalid amount. Use Scanner in main().

14. Key Takeaways

- private → most restricted.
- default → same package.
- protected → same package + inheritance.
- public → widest access.
- Access modifiers are important for encapsulation and data protection.